Lecture 2 — Probability Distributions & Statistical Hypothesis Testing

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IMG

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Part I — From data to distributions

Random variables

A random variable takes a value determined by the outcome of a random process.

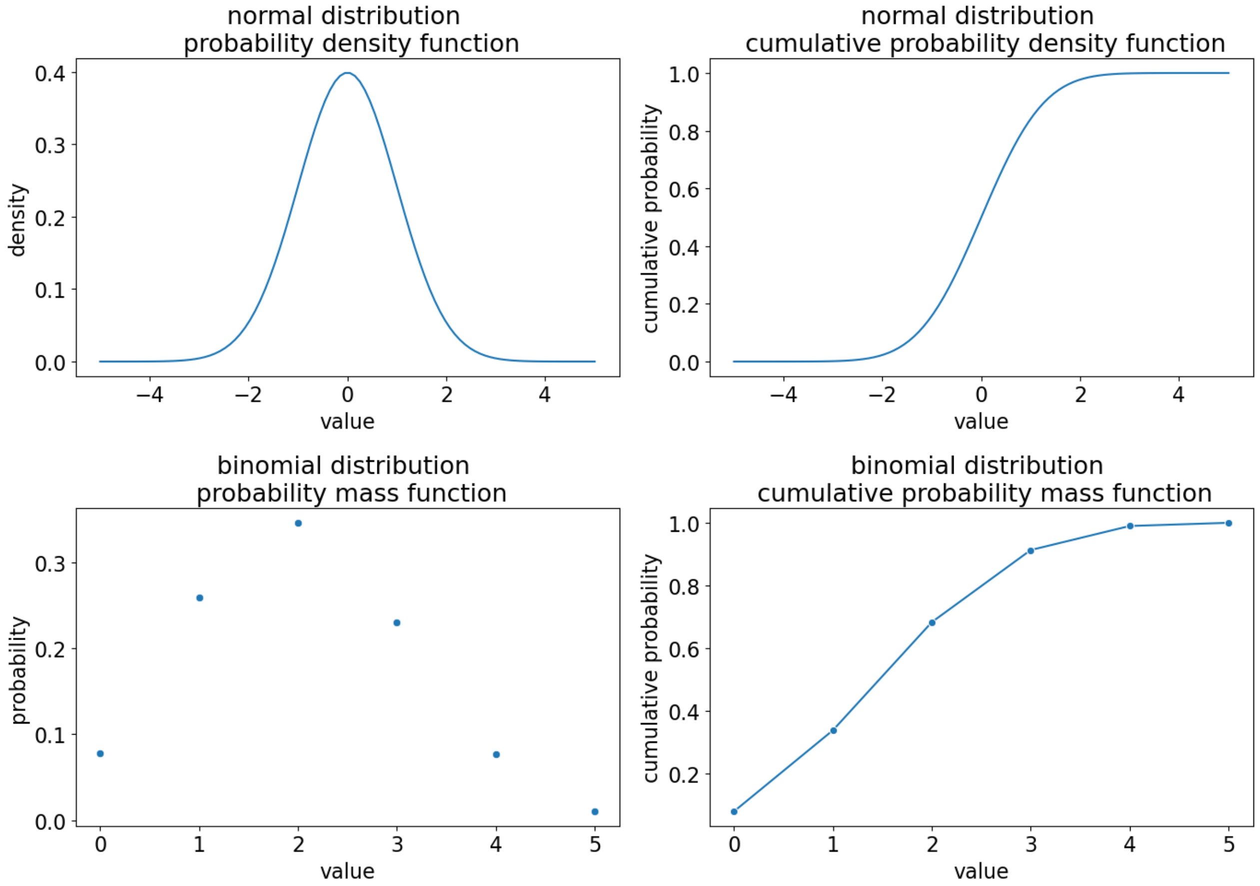
- A plain variable holds any value (x, y); a random variable (X, Y) gets its value from a random experiment — a coin toss: $X = 1$ heads, $X = 0$ tails.
- The Session 1 dataset was a stack of observed values of random variables; now we model the mechanism behind them.
- That model is a probability distribution — which values X can take, and how likely each is.

We described data; now we model the process that generated it.

Probability distributions: PMF, PDF, CDF

Three views of the same object.

- PMF (probability mass function) — discrete variables; the y-axis is a genuine probability, and the bars sum to 1.
- PDF (probability density function) — continuous variables; the y-axis is density, not probability. Probability = area under the curve; total area = 1.
- CDF (cumulative distribution function) — probability of a value ≤ x; runs from 0 to 1.

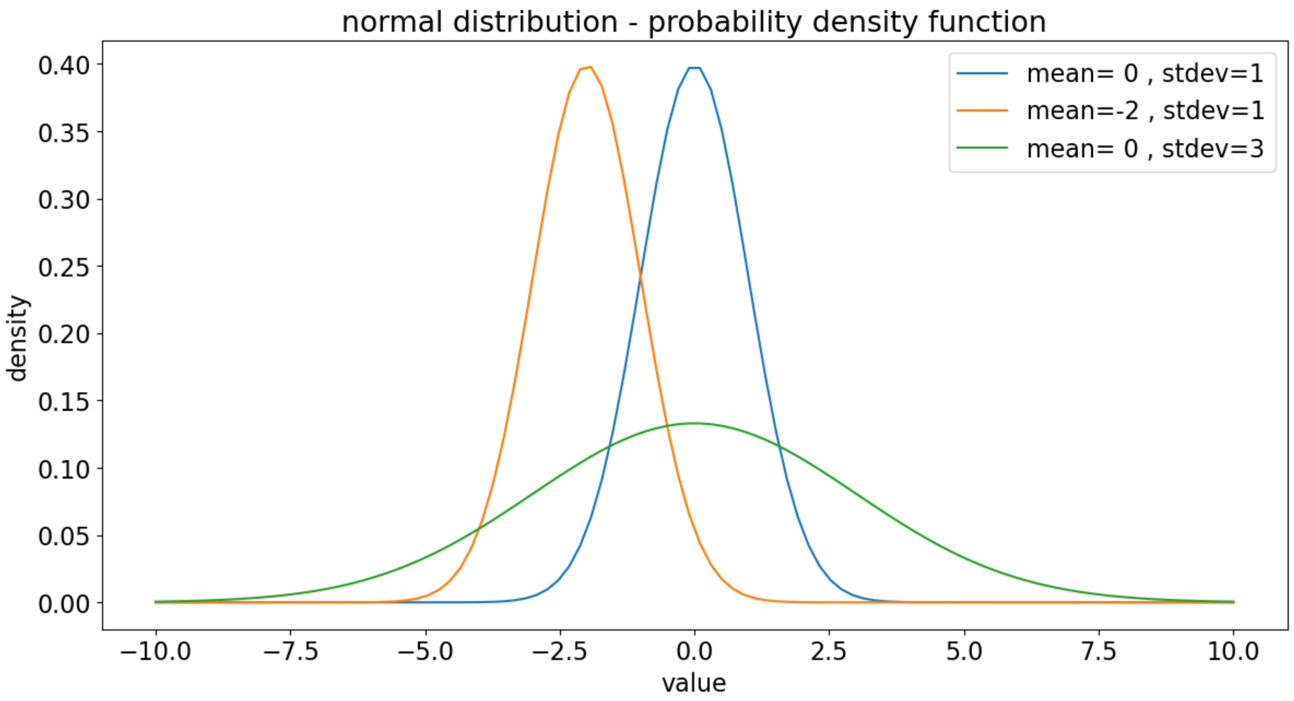


On a PDF it's area, not height, that gives the probability.

Parameters and notation

A distribution is pinned down by its parameters.

- Parameters set location, scale, and shape. The normal has mean μ (location) and standard deviation σ (scale).
- Notation: $X \sim N(\mu, \sigma)$ — “X follows a normal distribution with mean μ and standard deviation σ.”
- Changing a parameter slides, stretches, or reshapes the same family of curves.



One functional form, many shapes — set by the parameters.

Part II — From a sample to an estimate

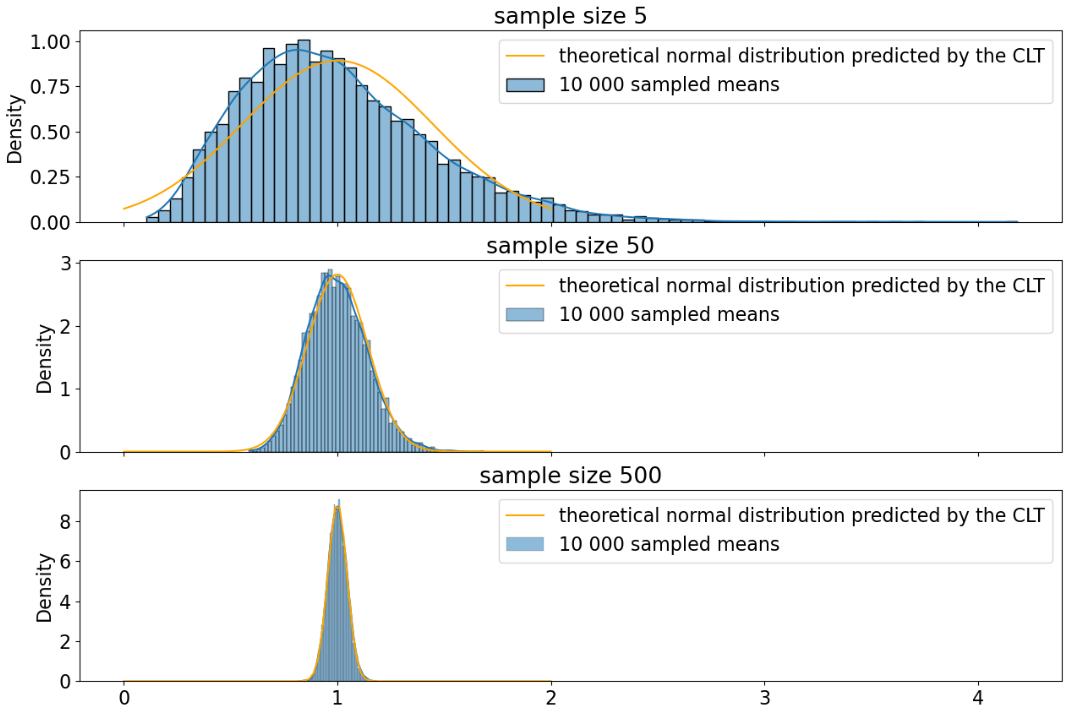
The Central Limit Theorem

The sample mean is itself a random variable — and it's nearly normal.

- Across many samples, the distribution of their means is approximately normal, even when the data are not.
- Result: the sample mean is approximately normal, with spread shrinking as the sample grows.

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- So we can reason about means without knowing the data's true distribution.

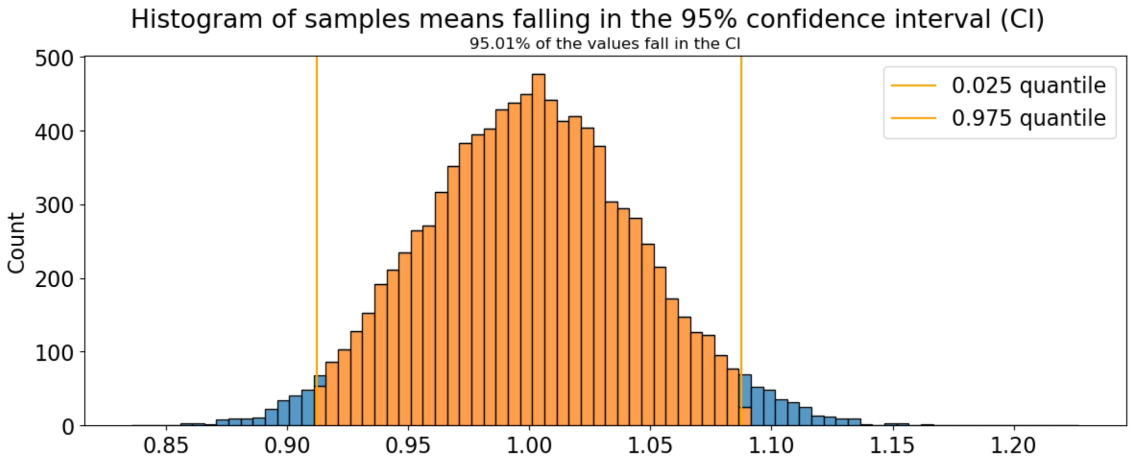


$\frac{\sigma}{\sqrt{n}}$ — *the SEM from Session 1, reappearing.*

Quantiles & confidence intervals

Turn a distribution into a range of plausible values.

- A quantile is the value below which a given fraction lies: 0.5 = median; 0.025 and 0.975 cut off the bottom and top 2.5%.
- A 95% confidence interval for the mean is bounded by the 0.025 and 0.975 quantiles of the sampling distribution.
- Over many repeated samples, 95% of such intervals contain the true mean.
- It quantifies how precisely the sample has located the true mean.



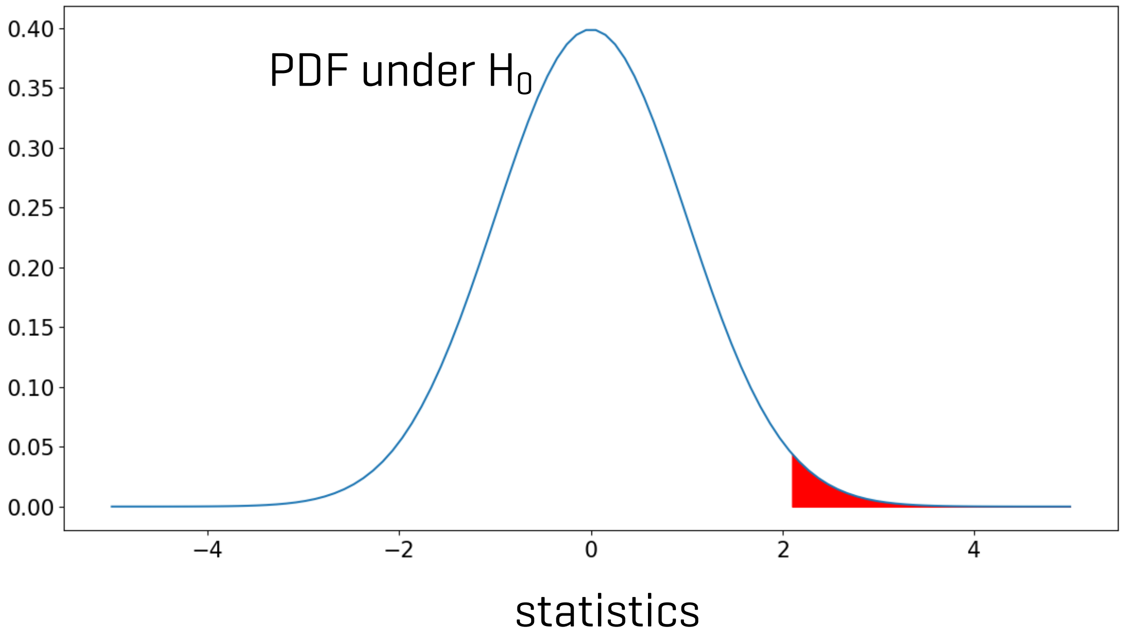
95% lives in the procedure, not in any single interval.

Part III — The logic of hypothesis testing

Anatomy of a hypothesis test

Four steps, always the same.

- State a null hypothesis H_0 and an alternative H_1 — claims about the real world.
- Pick a test statistic whose distribution under H_0 is known.
- Compute the statistic from your data.
- Ask how extreme it is relative to that null distribution.



Every test: a number whose behaviour we can predict if nothing is going on.

Motivation: is the true mean different from μ_0?

We have a sample; we want to judge the population mean behind it.

- Is the true mean μ different from a reference μ_0? (e.g. a batch's mean weight vs the labelled 500 g.)
- We only see the sample mean $\bar{x}$ — it almost never equals μ_0 exactly.
- Is the gap $\bar{x} - \mu_0$ real, or just sampling noise?
- The CLT gives the noise yardstick $\dfrac{\sigma}{\sqrt{n}}$
- Standardising the gap against it is the one-sample Student t-test.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \qquad \mathrm{df} = n - 1$$

$\bar{x} \neq \mu_0$ isn't enough — is the gap bigger than sampling noise?

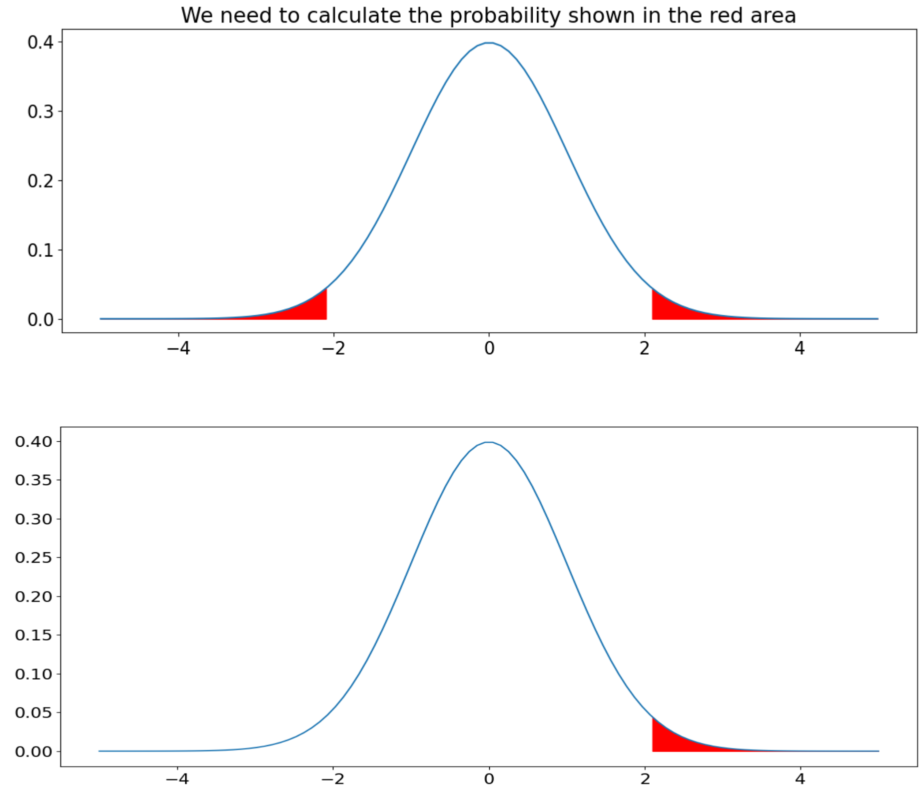
A worked test, one- vs two-sided

Centre and scale the sample mean.

- If σ is known, the centred-and-scaled sample mean is a standard normal under H_0: $\mu = \mu_0$ — easy to read probabilities off.

$$\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$

- Two-sided (H_1: $\mu \neq \mu_0$) counts extremes in both tails; one-sided (H_1: $\mu > \mu_0$) counts only one.
- For a symmetric statistic, the two-sided p-value is twice the one-sided.

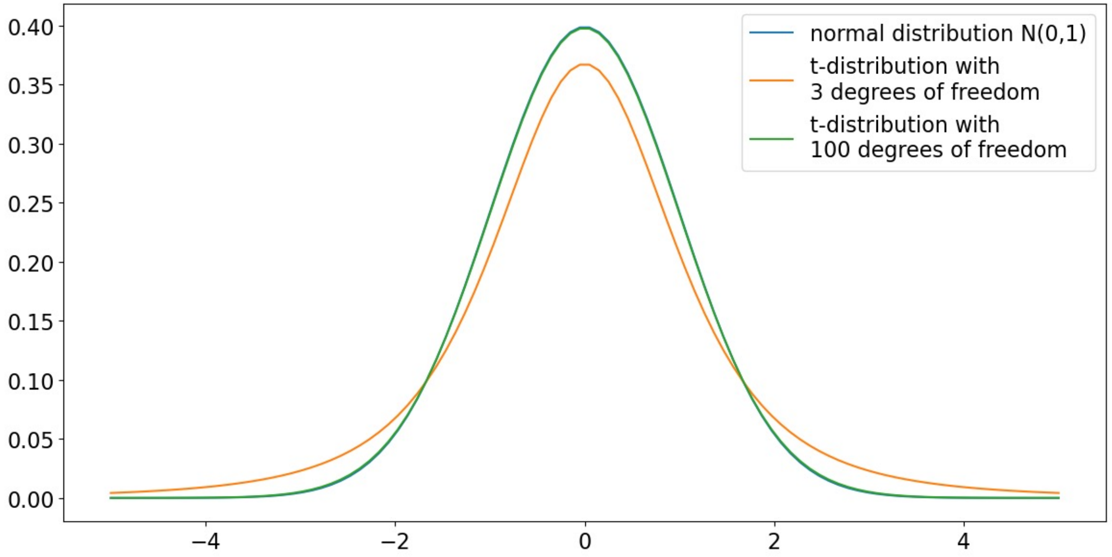


Commit to one- vs two-sided before seeing the data.

The t-distribution & degrees of freedom

We never really know σ — so we estimate it, And pay for it.

- Using the sample SD makes the statistic follow a t-distribution with $n-1$ DF, not a normal — fatter tails for the extra uncertainty.
- Degrees of freedom (DF): independent information left after estimating the mean — the same $n-1$ from Session 1.
- As DF grow, t converges to the normal (indistinguishable beyond ~100 DF).



Small samples → fatter tails → a bigger effect needed for significance.

Welch's t-test

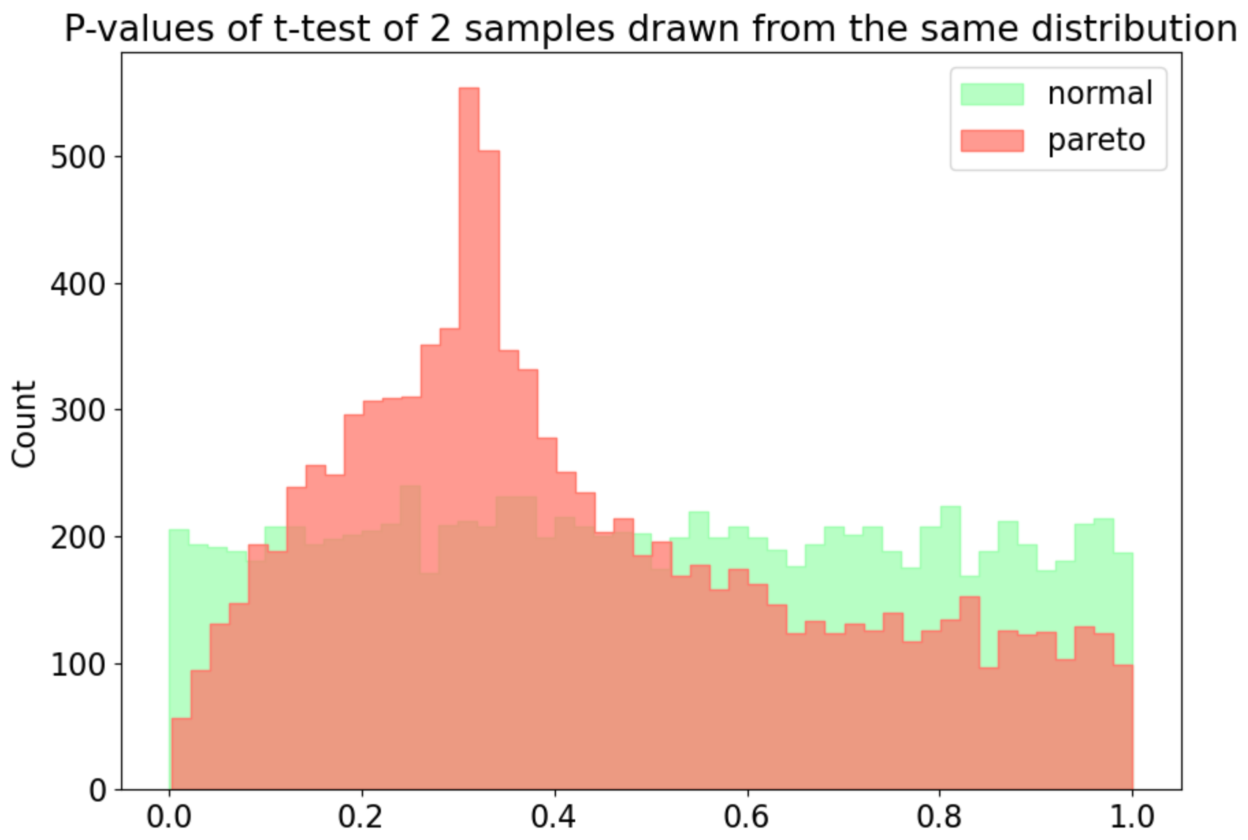
Comparing the means of two independent groups.

- H_0: $\mu_1 = \mu_2$ vs H_1: $\mu_1 \neq \mu_2$ — the test statistic is $$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

- Student vs. Welch: Student's assumes one pooled variance; Welch's uses each group's own variance + adjusted DF, so it needs no equal-variance assumption.
- Welch is the safe default
- a paired t-test fits matched observations (before/after).

When assumptions fail, the p-value stops meaning anything.

P-value distribution



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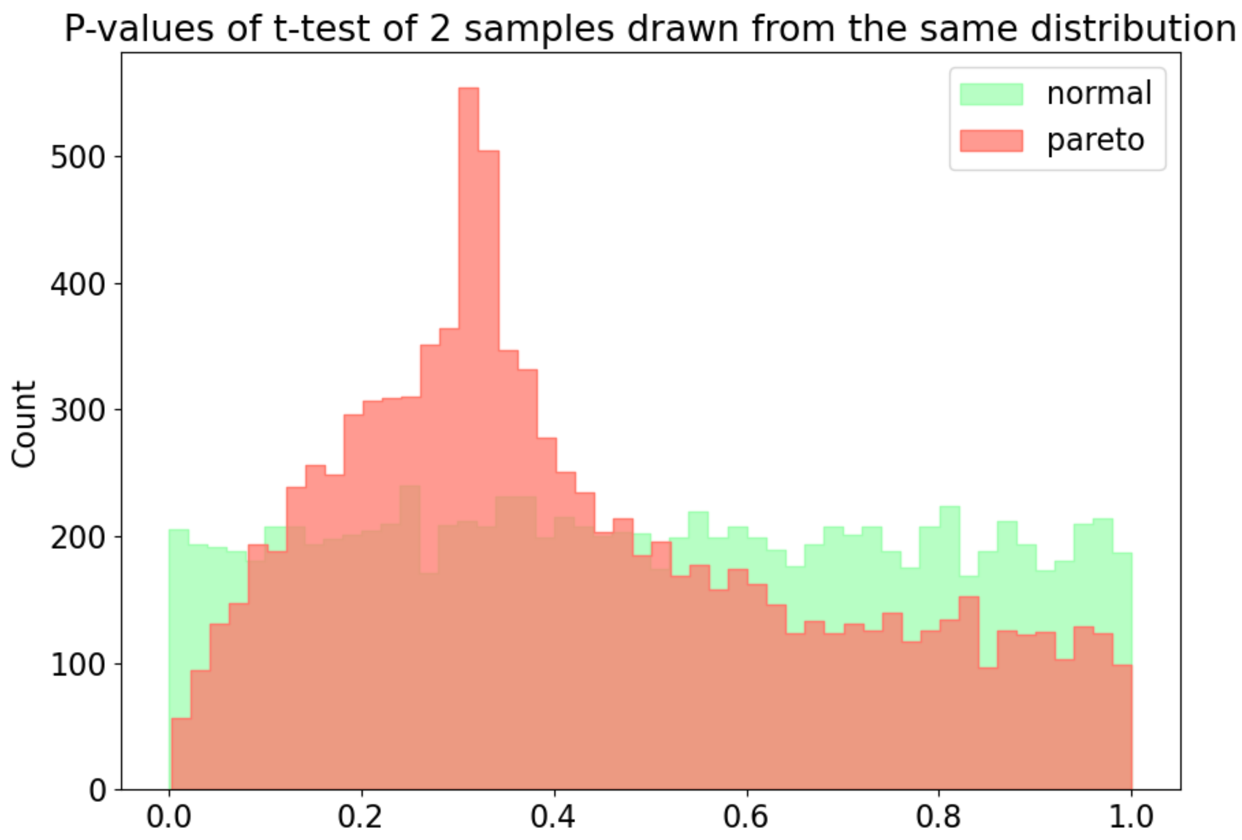
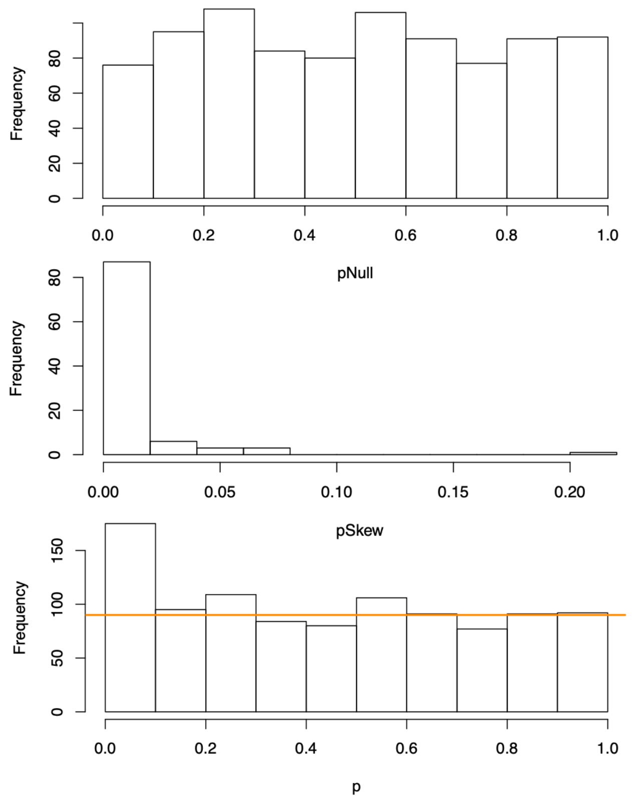




Null H_0 valid
uniform p-value distribution

Alternative valid
p-value distribution skewed to 0

All tests
p-value distribution is a mixture of uniform and skewed

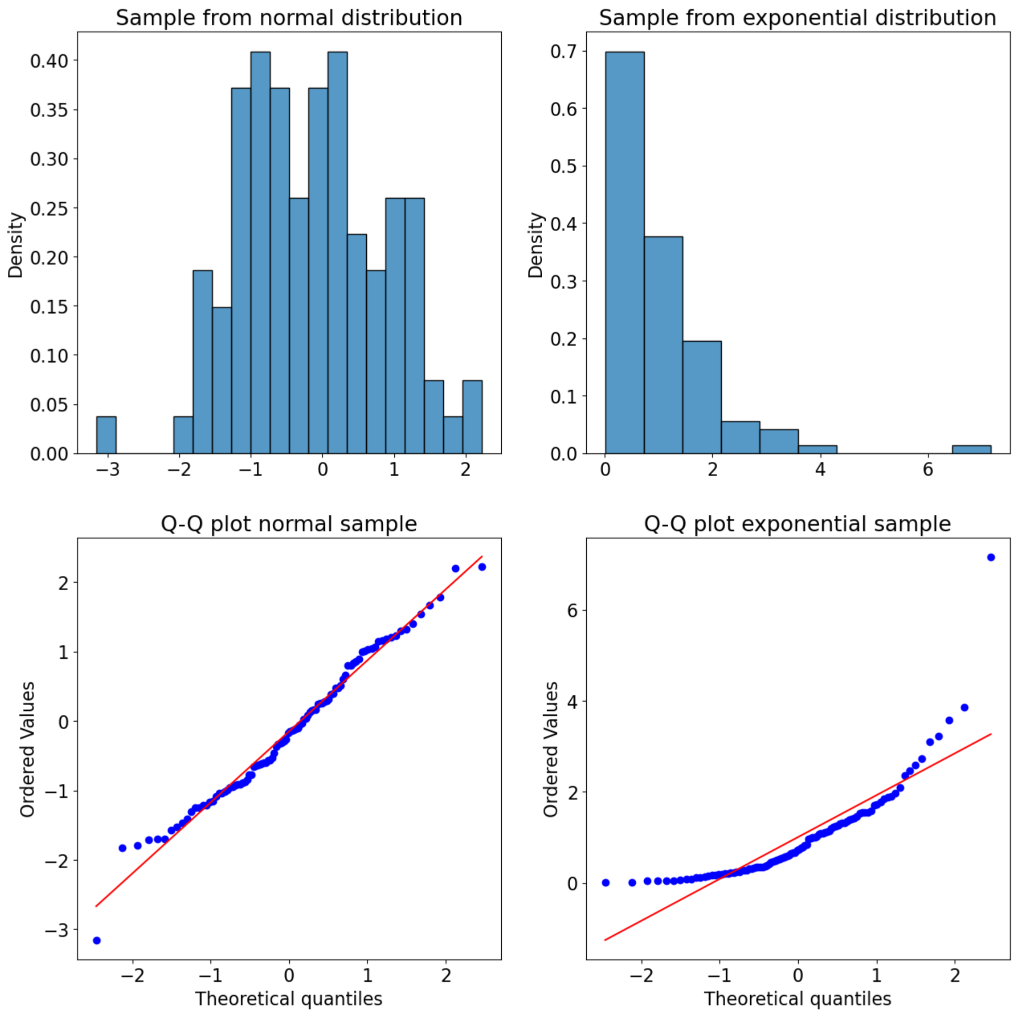
When assumptions fail, the p-value stops meaning anything.

Part IV — Checking assumptions & alternatives

Is it normal? Q–Q plots & Shapiro–Wilk

Check the assumption rather than hoping.

- Q–Q plot — sample vs theoretical normal quantiles; normal data fall on a straight diagonal, and the way points peel off shows how it deviates.
- Shapiro–Wilk — a formal test, H_0: "the data are normal"; a small p-value rejects normality.
- Best practice: use both — the picture and the test together.



Here a significant result is the bad news — non-normal.

When normality fails: Mann–Whitney U

A distribution-free alternative to the t-test.

- Nonparametric methods assume no distribution family — for when normality is genuinely violated and n is too small for the CLT.
- Mann–Whitney U works on ranks: H_0 — a random value from one group is equally likely above or below one from the other.

A different H_0 from the t-test — the two can legitimately disagree.

Part V — Reading results honestly

p-values and what they are not

The probability of data this extreme, if H_0 were true.

- Formally: P[statistic as or more extreme than observed | H_0 true]. Tied to a specific H_0.
- It is not: the probability H_0 is true, the probability of an effect, or your error rate.
- "Fail to reject H_0" is not "accept H_0."

Report the exact p-value — and pair it with an effect size.

Two ways to be wrong

Type I and Type II errors.

- Type I (false positive) — rejecting H_0 when it's true. Probability α, the threshold you set (often 0.05).
- Type II (false negative) — failing to reject H_0 when it's false. Probability β.
- They trade off: tightening α raises β, for fixed n.

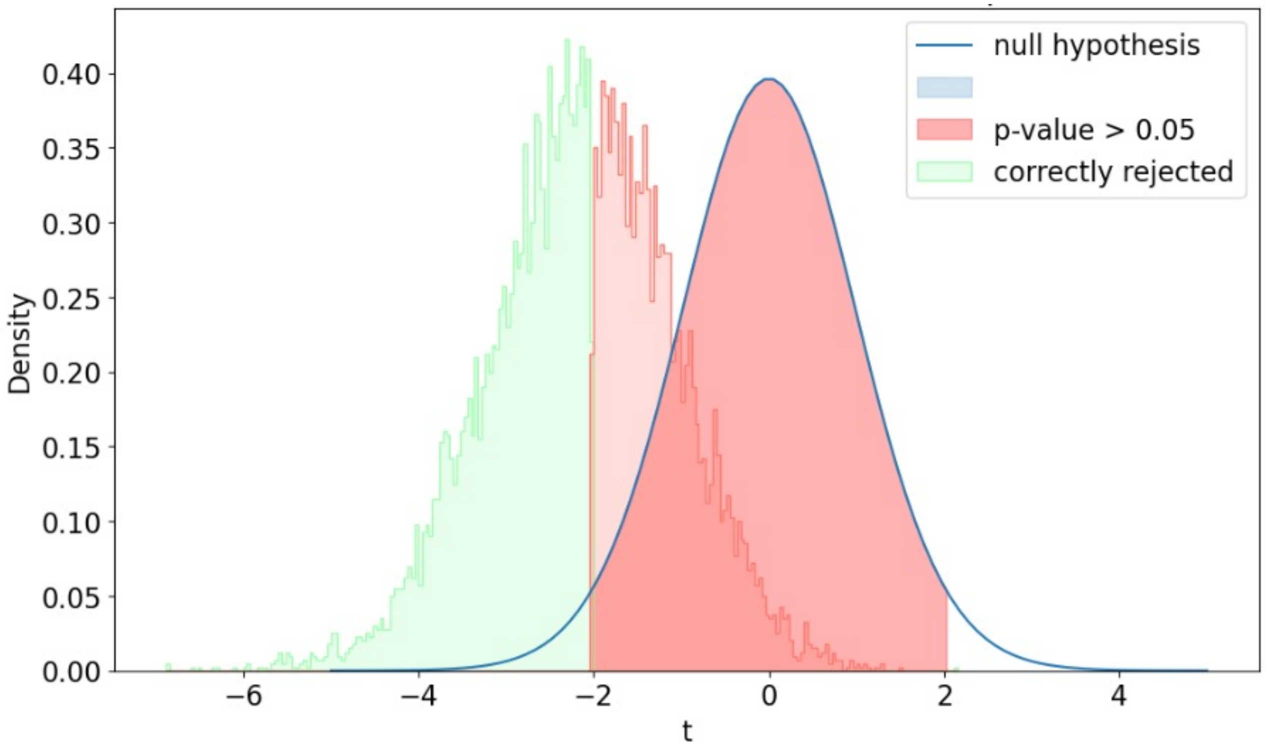
		REALITY	
		H_0 true	H_0 false
TEST	Accept H_0	**Correct acceptance** True negative	**Type II error** False negative probability = β
	Reject H_0	**Type I error** False positive probability = α	**Correct rejection** True positive power = $1 - \beta$

α is a choice, not a fact of nature; 0.05 is convention.

Power & effect size

Could the study even have detected the effect?

- Power = 1 − β — the chance of detecting a real effect; a design question, settled before data.
- It rises with sample size and true effect size; a study can "fail" simply by being underpowered.
- Effect size (Cohen's $d = \frac{|\mu_1 - \mu_2|}{\sigma}$) measures how big the difference is, independent of n.



Significance ≠ biological relevance — always report an effect size.

A field guide to common distributions

Match the distribution to the data-generating process.

Distribution	Generates	Parameters
Bernoulli	a single yes/no trial	p
Binomial	# successes in n trials	n, p
Multinomial	counts across > 2 categories	$n, p_1 \ldots p_k$
Hypergeometric	sampling without replacement	N, K, n
Poisson	# events in a fixed interval	λ
Geometric	# failures before first success	p
Negative binomial	# failures before r successes	r, p
Normal	symmetric continuous measurements	μ, σ
Log-normal	values whose log is normal	μ, σ of log
Chi-square	sums of squared normals	df

Ask what process generated the data — that picks the distribution.